Let, $m \in C$. By Defn. 7, $\exists n \in B$ s.t. $f(n)=m$
By Defn. 7, $\exists w \in A$ s.t. $f(g(w))=m$.
$\therefore g(w)=n$. f is defined for each element in B.
$\therefore$ Since f is a bijection from B to C, (by Defn. 8), each element of B has a corresponding mapping in A, ie. $\forall z \in B, \exists w \in A$ s.t. $g(w)=z$.
By Defn. 7, g is onto. □ **

35) Find an example of functions f and g such that $f \circ g$ is a bijection, but g is not onto and f is not one-to-one.

Solution: $g: \{1,2\} \longrightarrow \{1,2,3\}$ $g(1)=2$ $g(2)=3$.
The co-domain has an element 1 which has no mapping in the domain of g. By Defn. 7, g is not onto.
$f: \{1,2,3\} \longrightarrow \{1,2\}$ $f(1)=f(3)=1$ $f(2)=2$
By Defn. 5, f is not one-to-one.
By Defn. 10, $f \circ g$ is the function from $\{1,2\}$ to $\{1,2\}$
$f(g(1))=f(2)=2$ $f(g(2))=f(3)=1$.
It's easy to see that from Defn. 5 and Defn. 7, $f \circ g$ is both one-to-one and onto. By Defn. 8, $f \circ g$ is a bijection. II

36) If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer.

Solution: Let, $f: B \to C$ and $g: A \to B$. By Defn. 10, $f \circ g: A \to C$
By Defn. 5, $\forall x \forall y, x, y \in B$ if $f(x)=f(y)$, then $x=y$.
By Defn. 5, $\forall z \forall w, z, w \in A$, if $f(g(z))=f(g(w))$, then $z=w$.
Suppose, g is not one-to-one. By Defn. 5, $\exists a, b$ where $a, b \in A$ and $a \neq b$ and $g(a)=g(b)$. Let, $g(a)=g(b)=c$.
$f(c)=f(c) \Rightarrow f(g(a))=f(g(b))$. By Defn. 5, $a=b$. But this is a contradiction. $\therefore g$ must be a one-to-one function. □